Core Concepts of Pulse

Was this page helpful?Â

All tasks you perform with Pulse rely on a set of concepts that support the core of Pulse's functionality.

This page gives you the basics about the most important concepts of Pulse. For more details or for other concepts read the pages later in this chapter.

Data science loopâ

The data science loop is the process of creating and testing rules and models.

Data sourcesâ

Data sources are historical datasets imported to a data science environment. You can use them for data exploration, feature engineering, model training, rule testing, etc.

Plansâ

Plans define a transformation flow of one data source into another data source. They are useful to systematize data cleaning, data processing, and feature engineering, both in a data science pipeline and in runtime.

Modelsâ

Pulse enables you to quickly train machine learning models and evaluate the results over several iterations. It provides tools for you to extract relevant performance metrics and track your progress along each iteration of the data science loop.

Rulesâ

Rules define a pattern based on the fields of a data source that produces a binary result for each record that matches a condition. Pulse lets you define and evaluate the performance of blocks of rules using a data science approach.

Workflowsâ

Workflows represent the life cycle of a real-time event through the system. A workflow maps other concepts, such as data science concepts, into runtime components and ensures they are properly translated and orchestrated.

Listsâ

A list represents a set of values and aims at improving the efficiency of rules when the same pattern needs to be checked against a large number of entries. For example, a list of credit card numbers, a list of merchants, etc.

User-Defined functionsâ

User-defined functions (UDFs) enable you to extend Pulse beyond its off-the-shelf capabilities. They are especially useful when you need custom logic to process data, such as transforming data source fields with Plans or configuring conditions for rules.